

What is your counterparty's promise to pay worth?

CVA and DVA with simulation
and machine learning

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Sample class lecture notes

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Master's in Risk Management

Every numerical result is reproducible with the notebook that
accompanies these notes. Market data: Valmer.

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1 The problem: two identical swaps that are not worth the same

Picture two transactions that are exactly alike: a five-year interest rate swap, a notional of 10 million pesos, referencing TIIE de Fondo. The only difference is who signs on the other side. One is with a top-tier bank; the other, with a company whose credit the market considers fragile.

The contractual cash flows are identical. And yet the two transactions are **not worth the same**, because in the second one there is a higher probability that the counterparty will fail to pay what it owes when the time comes. That difference in value has a name: the **credit valuation adjustment**, or CVA.

The 2008 financial crisis turned this adjustment into standard practice. A recurring finding in post-crisis reviews is that the majority of derivative losses did not come from actual defaults, but from the **deterioration in the market value of the promise to pay**: counterparties that were still very much alive, but whose credit had degraded so far that the value of their obligations collapsed. Anyone who had not already recognised that adjustment on the books had to recognise it all at once.

Why this is a profession, not an exam topic

CVA is now a global requirement —IFRS 13 and the Basel framework— and in Mexico it is part of routine regulatory reporting to the CNBV and to Banco de México (section 8). Banks and corporates with derivative portfolios maintain teams dedicated exclusively to computing and hedging this number; the industry calls them **XVA desks**. Whoever masters this machinery has a concrete job waiting.

These notes develop the class in more depth than forty minutes allow. They assume familiarity with probability, present value and the basic mechanics of a swap, but not with stochastic calculus: every result that is needed is stated and explained, though not always proved.

2 The formula, in three pieces

The CVA of a transaction maturing at T is written

$$\text{CVA} = \underbrace{(1 - R)}_{\text{severity}} \int_0^T \underbrace{\text{EE}(t)}_{\text{exposure}} \underbrace{d\text{PD}(t)}_{\text{probability}} . \quad (1)$$

The expression is far less intimidating when read as three common-sense questions, one per factor.

Severity: how much do I lose if it defaults? The factor $(1 - R)$ is the loss given default, where R is the recovery rate: the fraction of the amount owed that is recovered in bankruptcy proceedings. In emerging markets the usual convention is $R = 0.25$; a quarter is recovered and three quarters are lost.

Exposure: how much will it owe me at each future date? The term $\text{EE}(t)$ is the expected exposure at time t : the average of what the counterparty would owe me if it defaulted at that moment. This is the hard piece, because the future value of a swap depends on interest rates that do not yet exist. Section 4 is devoted to it.

Probability: how likely is it to default? The differential $d\text{PD}(t)$ is the probability that default occurs right around t . It is read off the credit derivatives market, as section 3 explains. In words: CVA is the average, across all future dates, of *what I would lose* multiplied by *the*

probability of losing it. It is the price of an insurance policy nobody sold, but which the market charges for anyway.

2.1 Why market probabilities and not historical ones

A natural question is why not use default probabilities estimated from historical data —observed default rates, rating transition matrices— instead of those implied by the credit market. The answer separates two activities that look alike but are not the same.

Forecasting answers how likely it is, in the real world, that this firm defaults. That is a statistical question, answered with historical data, under what the theory calls the physical measure $\mathbb{P}$.

Pricing answers how much it costs today to protect against that default. That is a price question, and the market supplies the answer: if an instrument exists that pays out when the counterparty defaults —a CDS— its quote contains all the information required. This is risk-neutral valuation, under the measure $\mathbb{Q}$.

The distinction people get wrong most often

CVA is a **price**, not a forecast: it is the discount applied to the value of a transaction, and it must be replicable by buying credit protection. That is why it is computed under $\mathbb{Q}$. CDS-implied probabilities are typically higher than historical ones because they embed the premium investors demand for bearing risk; that premium is part of the price and must not be stripped out.

The insurance analogy is best avoided. Insurance rests on risk mutualisation and is priced under $\mathbb{P}$; CVA rests on replication and is priced under $\mathbb{Q}$. Conflating them leads to wrong conclusions.

3 The probability is read off the market

A credit default swap (CDS) is a contract in which one party pays a periodic premium and, in exchange, receives compensation if a given issuer defaults. Its quote —the spread, in annual basis points— is literally the market price of that protection.

Definition 3.1 (Hazard rate). Default is modelled as the first jump of a counting process with intensity λ , called the hazard rate. Under that model, the probability of surviving to t is

$$Q(t) = e^{-\lambda t}. \quad (2)$$

The link to the market comes from equating, at inception, the present value of the CDS premiums with the present value of the expected compensation. Under a constant intensity that equality yields the approximate relation

$$\lambda \approx \frac{s}{1 - R}, \quad (3)$$

where s is the spread in decimal form.

Remark 3.2 (The intuition behind the ratio). If protection costs s per year and each default costs $(1 - R)$, then the implied default frequency is the ratio of the two. A spread of 150 basis points with a 25% recovery implies $\lambda = 0.015/0.75 = 2\%$ per year.

The probability of defaulting within an interval $[t_{i-1}, t_i]$ —what (1) denotes dPD — is simply

the drop in the survival curve:

$$\Delta PD(t_i) = Q(t_{i-1}) - Q(t_i). \quad (4)$$

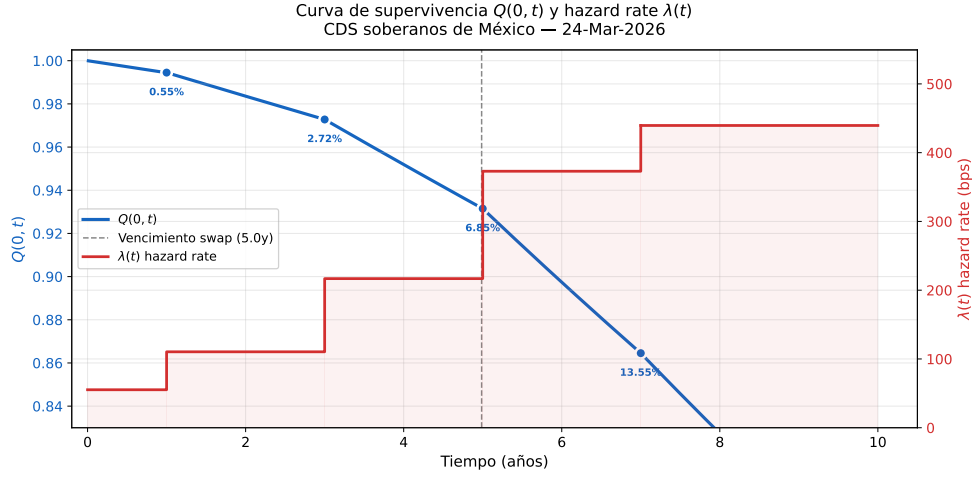


Figure 1: From the CDS spread to the survival curve. The bootstrap translates market quotes into default probabilities by tenor.

In practice the spread is not a single number: it is quoted at several tenors and the intensity is extracted piecewise, fitting each tenor sequentially so that all quotes are reproduced. That procedure is known as bootstrapping the credit curve. When the counterparty has no quoted CDS —the case for most counterparties in Mexico— a proxy curve is used: typically the sovereign curve, scaled by sector and credit quality. It is an approximation, and it is worth saying so.

4 Exposure: the future has to be simulated

To know how much the counterparty will owe me two years from now, I have to know what the swap will be worth two years from now, and that depends on the interest rates prevailing then. There is no way around it: scenarios of future rates have to be generated.

4.1 How many factors the curve needs

Before picking a model it pays to ask the data how many sources of uncertainty actually drive the curve. The standard tool is principal component analysis (PCA) applied to daily changes in rates across tenors.

Component	Variance explained	Interpretation
PC1	79.85%	Level: the whole curve moves up or down together
PC2	10.38%	Slope: short and long ends move in opposite directions
PC3	≈4%	Curvature: the belly moves relative to the wings

The reading is direct. A single-factor model —Vasicek, Hull-White— can only generate level moves: if the short rate rises, *every* rate rises. Such a model imposes perfect correlation across tenors and cannot produce steepenings, flattenings or inversions.

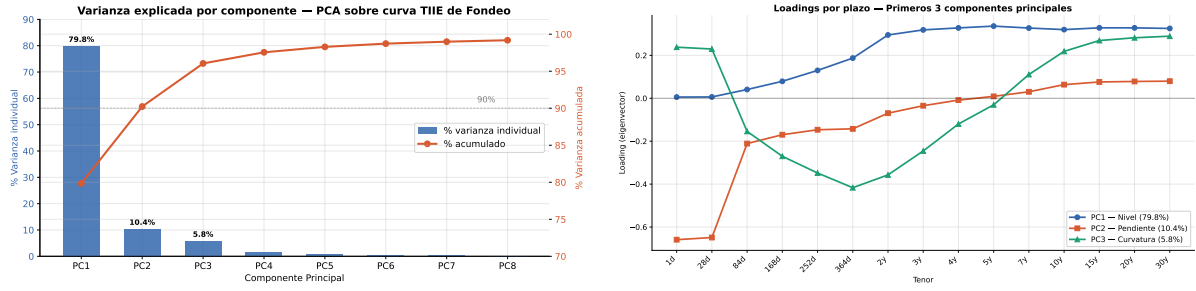


Figure 2: PCA of the TIIE de Fondeo curve. Left: variance explained by component. Right: loadings by tenor. The first component is roughly flat (level); the second changes sign between the short and the long end (slope).

Why it matters

For CVA this is not a cosmetic detail. The value of a swap depends on the shape of the curve, not just its level; a model that cannot deform the curve understates the variability of exposure and, therefore, the CVA. Since level and slope together explain roughly 90% of the variance, **two factors are necessary and sufficient** for this purpose.

4.2 The G2++ model

The G2++ model (Brigo and Mercurio, 2006) decomposes the instantaneous short rate into two correlated Gaussian factors plus a deterministic shift:

$$r(t) = x(t) + y(t) + \varphi(t), \quad (5)$$

with dynamics under the measure $\mathbb{Q}$

$$dx(t) = -a x(t) dt + \sigma dW_1(t), \quad x(0) = 0, \quad (6)$$

$$dy(t) = -b y(t) dt + \eta dW_2(t), \quad y(0) = 0, \quad (7)$$

$$dW_1(t) dW_2(t) = \rho dt. \quad (8)$$

Each factor is an Ornstein-Uhlenbeck process: it reverts to zero at speed a (or b) and receives random shocks of magnitude σ (or η). The shift $\varphi(t)$ is deterministic and is calibrated so that the model reproduces **exactly** the curve observed today; without it, the model would produce prices inconsistent with the market from day one.

The link between the PCA and the two factors

The mean reversion speed determines how each factor affects different tenors.

A slow factor (small mean reversion speed) persists over time, so its shocks affect rates at all tenors alike: this is the **level** factor, the analogue of PC1.

A fast factor (large mean reversion speed) dies out before reaching the long tenors, so it moves the short end and leaves the long end still: it deforms the **slope**, the analogue of PC2.

The parameters calibrated to ATM swaptions on the TIIE de Fondeo curve, with a reference date of 24 March 2026, are:

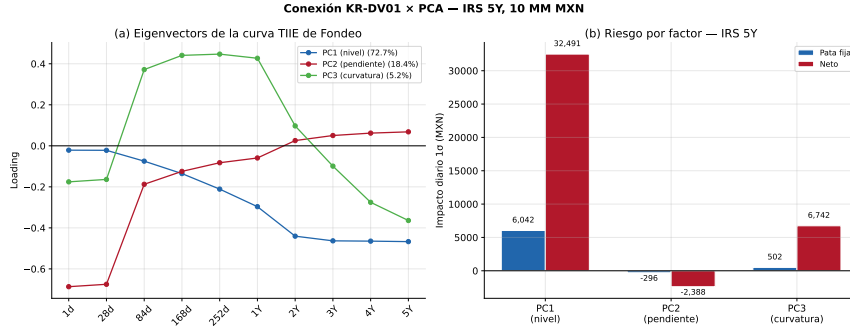


Figure 3: Correspondence between the empirical PCA loadings and the sensitivity functions of the two G2++ factors.

Parameter	Value	Role
a	0.148849	mean reversion of the slow factor (level)
b	0.982642	mean reversion of the fast factor (slope)
σ	0.026973	volatility of the slow factor
η	0.062209	volatility of the fast factor
ρ	-0.913918	correlation between factors

Remark 4.1 (Reading the parameters). The fast factor reverts almost seven times more quickly than the slow one, exactly as the PCA anticipated. The strongly negative correlation says that when the level rises, the curve tends to flatten: a well-documented feature of the Mexican market.

4.3 Exact simulation

A practical virtue of the G2++ is that its factors admit **exact simulation**. Since each factor is Gaussian, its conditional distribution is known in closed form and it suffices to draw normals with the right mean and variance:

$$x(t + \Delta) = x(t) e^{-a\Delta} + \sqrt{\frac{\sigma^2}{2a} (1 - e^{-2a\Delta})} Z_1, \quad (9)$$

$$y(t + \Delta) = y(t) e^{-b\Delta} + \sqrt{\frac{\eta^2}{2b} (1 - e^{-2b\Delta})} Z_2, \quad (10)$$

where Z_1 and Z_2 are standard normals with correlation ρ , obtained from two independent normals via $Z_1 = \varepsilon_1$ and $Z_2 = \rho \varepsilon_1 + \sqrt{1 - \rho^2} \varepsilon_2$.

Proposition 4.2 (No discretisation error). The difference from an Euler-type scheme is substantial. Euler introduces an error proportional to $\sqrt{\Delta}$, which forces small time steps. Here **there is no discretisation error, whatever the step size**: the only error is the Monte Carlo one, arising from a finite number of scenarios and decaying as $O(1/\sqrt{N})$. This makes it possible to align the time grid with the swap's payment dates without sacrificing accuracy.

Once the factors are simulated, the price of any zero-coupon bond in any scenario follows in closed form:

$$P(t, T) = A(t, T) \exp(-B_1(t, T) x(t) - B_2(t, T) y(t)), \quad (11)$$

where $B_1(t, T) = (1 - e^{-a(T-t)})/a$ and likewise B_2 with b , while $A(t, T)$ collects the fit to the initial curve and a convexity correction. With this formula the full curve is rebuilt at every node of every scenario, and the swap is priced off it.

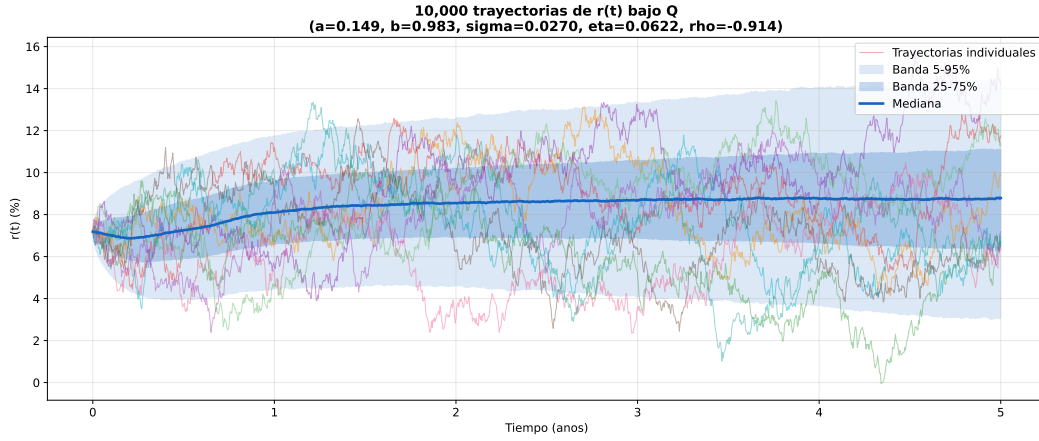


Figure 4: Ten thousand short rate paths under $\mathbb{Q}$ with the calibrated parameters. The fan widens with the square root of time: this is not a rate forecast, but the uncertainty the market is pricing today.

5 From scenarios to the hump

Pricing the swap in every scenario and at every date yields a distribution of values per date. Three measures are extracted from it:

$$EE(t) = \mathbb{E}[\max(V(t), 0)], \quad (12)$$

$$ENE(t) = \mathbb{E}[\max(-V(t), 0)]. \quad (13)$$

Expected exposure $EE(t)$ averages only the positive part: if the swap is in my favour and the counterparty defaults, I lose that value; if it is against me, I lose nothing because I still owe. Hence the $\max(\cdot, 0)$. **Expected negative exposure** $ENE(t)$ is the mirror image. **Potential future exposure** (PFE) is a high percentile —usually 97.5%— and is used to set credit lines, not to price.

5.1 The common origin of CVA and DVA

At any future date, the value of the swap across scenarios forms a distribution that is roughly symmetric around zero if the trade was struck at the money. That single distribution, split at zero, generates both exposures.

Two risks, one simulation

The **positive side** is what the counterparty owes me. If it defaults, I lose. Its average is $EE(t)$, and it feeds the **CVA**.

The **negative side** is what I owe the counterparty. If I default, it loses. Its average is $ENE(t)$, and it feeds the **DVA**.

These are not two separate calculations and they do not require two models: they are two readings of the same simulation. For an at-the-money trade both exposures are nearly identical, so all the asymmetry in the final price comes from each party's credit spread.

Remark 5.1 (The non-linearity that complicates everything). The $\max(\cdot, 0)$ operator in (12) deserves a note, because almost every technical difficulty in this field is born there. Being non-linear, it makes CVA **non-additive**: the CVA of a portfolio is not the sum of the individual CVAs. That is precisely the rationale for netting, and also the reason why computing the effect of adding a new trade to an existing portfolio is a problem in its own right.

5.2 Why exposure is hump-shaped

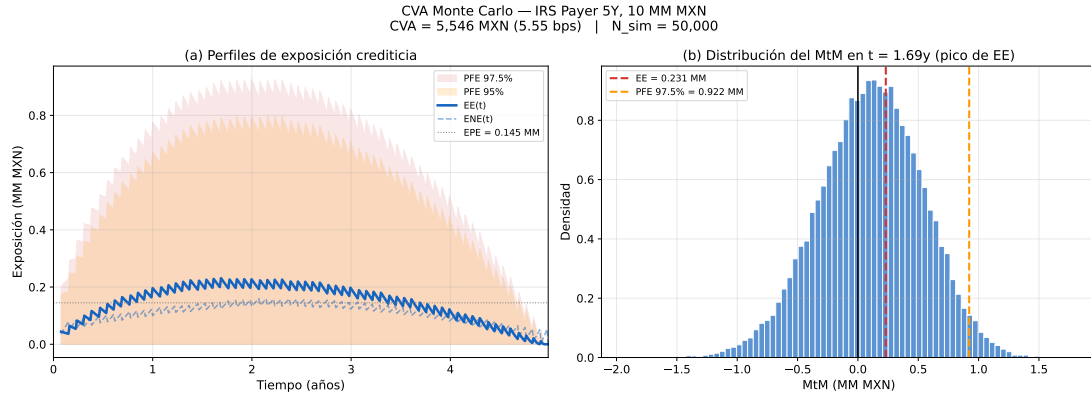


Figure 5: Exposure profiles of the reference swap. EE rises through rate diffusion and falls through amortisation; the peak appears around $T/3$.

The profile has a characteristic shape—it rises, peaks and decays to zero—that results from two opposing forces.

Diffusion pushes it up. The further out you look, the more uncertainty has accumulated about future rates, so the swap can drift further from zero in either direction. Dispersion grows roughly with $\sqrt{t}$.

Amortisation pushes it down. As time passes there are fewer cash flows left to exchange, so even if rates move a great deal, there is less to gain or lose. At maturity nothing remains: exposure is necessarily zero.

The maximum appears where the two forces balance, roughly at $T/3$ for a fixed-rate swap. For the five-year reference swap, the peak occurs around $t = 1.7$ years.

The operational conclusion

Counterparty risk does not peak at maturity, but in the first third of the trade's life. That is where credit limit monitoring should be concentrated. It is one of those conclusions intuition places in the wrong spot.

6 The price: CVA, DVA and bilateral netting

With both pieces in hand, CVA is computed by discretising (1) over the time grid:

$$\text{CVA} = (1 - R) \sum_i \text{EE}(t_i) \Delta \text{PD}(t_i) \text{DF}(t_i), \quad (14)$$

where $\text{DF}(t_i)$ is the discount factor. DVA is the same expression with the two substitutions already anticipated—the other profile and the other credit:

$$\text{DVA} = (1 - R) \sum_i \text{ENE}(t_i) \Delta \text{PD}^{\text{own}}(t_i) \text{DF}(t_i). \quad (15)$$

The bilateral adjustment is the difference between the two:

$$\text{BCVA} = \text{DVA} - \text{CVA}. \quad (16)$$

Example 6.1 (The reference swap). For the five-year payer swap on 10 million pesos, with the counterparty quoted at 150 basis points and ourselves at 80, the calculation gives $\text{CVA} = 7.65$

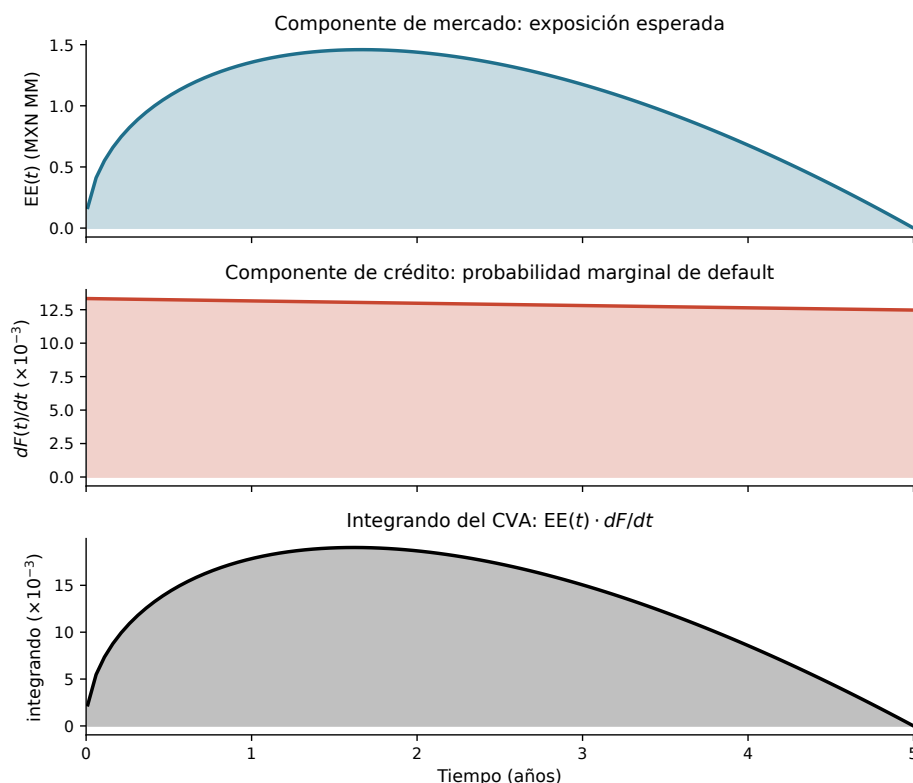


Figure 6: Anatomy of the integrand: expected exposure (top), marginal default probability (middle) and their discounted product (bottom). The area under the last curve is the CVA.

bps and DVA = 4.21 bps, hence BCVA = −3.44 bps: some 3,400 pesos coming straight out of the margin on the trade. The counterparty is the riskier of the two, so the net adjustment costs us.

Here is the answer to the question that always comes up in class: **yes, they net**, and the sign of the result depends on which of the two is riskier. If my counterparty trades wider than I do, CVA dominates and the adjustment costs me. If I trade wider than it does, DVA dominates and the adjustment works in my favour.

Proposition 6.2 (Break-even point). The bilateral adjustment vanishes when both spreads coincide: two equally risky counterparties charge each other the same. When two institutions negotiate the price of a swap, much of the discussion is precisely about this difference.

6.1 The awkwardness of DVA

DVA has a property worth stating plainly: **it is an accounting gain that increases as your own credit deteriorates**. If my institution weakens, my liabilities are worth less and that is booked as profit.

The reasoning is internally consistent—it reflects that my liabilities are worth less in the market—and that is why it is admissible under the fair value criterion of IFRS 13. But that gain can only be monetised by defaulting, which makes it useless as a solvency buffer. For that reason the Basel framework excludes own credit from regulatory CVA capital, and for that same reason the thesis behind this class computes **unilateral CVA**: DVA appears here to complete the conceptual picture, not as a result of the work.

7 The operational problem and machine learning

With the engine described above, pricing the CVA of one swap takes on the order of 264 milliseconds. That sounds fast, until it is multiplied by the reality of a trading desk:

- A book of 795 trades requires some 3.5 minutes per full valuation.
- Computing sensitivities means repeating the calculation with the curve and the spreads bumped in dozens of directions: around 2.3 hours.
- A stress exercise with a thousand scenarios is measured in days of compute.

And it is not only about periodic reporting. The desk needs answers **during the day**: how much CVA is added by a trade being quoted right now, how to hedge the book, which counterparty the number comes from. At that scale, an engine taking hundreds of milliseconds per trade stops being viable.

7.1 Learning from the engine itself

The idea is to train a statistical model that learns the function mapping the features of a trade and of the market to its CVA. What makes the approach viable is where the training data comes from: **the Monte Carlo engine generates it**. There is no manual labelling and no external data; the teacher is the simulator.

Simulate once, price everything

The architecture is cheaper than it first appears. Since the cube of simulated paths can be reused to reprice the same swap under different credit spreads or different fixed rates, thousands of training examples cost a single simulation. That principle is what brings dataset generation down from hours to seconds.

Two model families are natural here. **Gradient boosted trees** (XGBoost) are hundreds of decision trees in which each one corrects the errors of the previous: the standard for tabular data, with good interpretability. **Neural networks** are compositions of linear transformations and non-linear activations; they produce a smooth function, which —as we shall see— matters.

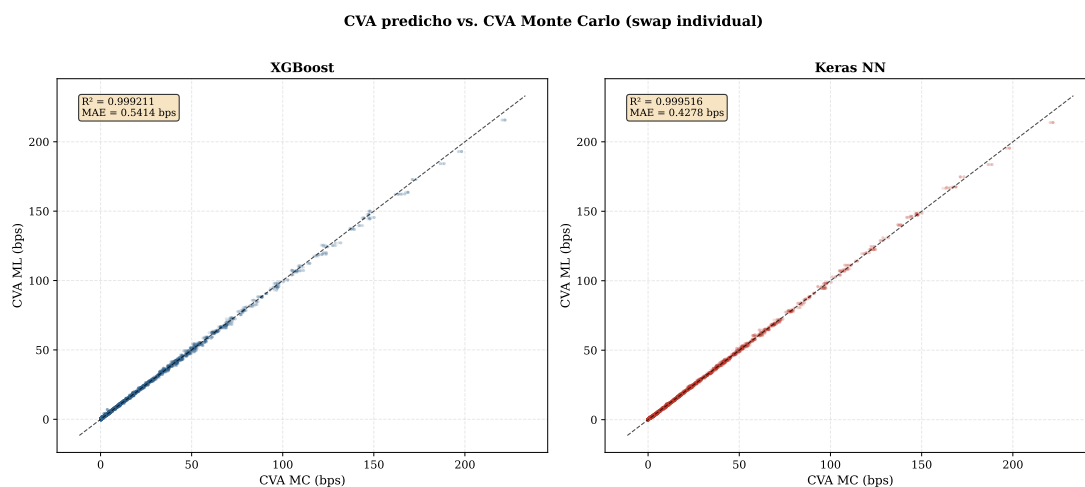


Figure 7: Predicted CVA against Monte Carlo CVA for both models. Each point is a swap from the test set; the dashed line is the identity.

Method	Time per swap	Speed-up	R ²
Monte Carlo	264 ms	1×	benchmark
Analytical approximation	142 ms	≈2×	—
XGBoost	1.7 μ s	≈155,000×	0.99921
Neural network	50.1 μ s	≈5,300×	0.99952

7.2 How to read the speed-up honestly

A figure like 155,000× demands precision about what it measures, because it is easy to oversell. It is **marginal query latency**: the time to answer one additional question with a model already trained and resident in memory. It excludes generating the labels, training, validating and periodic retraining.

Over the full lifecycle, preparation costs on the order of minutes and each query saves hundreds of milliseconds. The break-even point arrives after a few thousand queries—a volume a desk covers in a single day of quoting. For an isolated query, by contrast, the Monte Carlo engine remains cheaper end to end. The speed-up is real **in the repeated-use regime**, which is exactly a trading desk’s regime.

The surrogate does not replace the engine

The engine generates the labels, validates the surrogate and signs the auditable report. What the surrogate enables is not doing faster what was already being done, but doing what could not be done before: exploring thousands of scenarios in the time it takes to hold a conversation with a client.

7.3 A result worth knowing: the model depends on the question

Trees predict the *level* of CVA with remarkable accuracy. However, if one tries to obtain a *sensitivity*—the derivative of CVA with respect to the curve or the spread—by differentiating that same model trained on levels, the result is unusable.

Remark 7.1 (The reason is structural). A tree is a step function: constant within each leaf, with jumps at the boundaries. Its derivative is zero almost everywhere and undefined at the jumps. That fingerprint is visible to the naked eye in a CVA heat map generated with trees: the contour lines are staircases, not smooth curves.

The conclusion is not that trees are bad, but that **the right model depends on what you will ask of it**. For levels, trees are excellent. For sensitivities obtained by differentiation a smooth model is preferable—a regularised linear regression or a neural network—whose derivative exists and is continuous. A tree trained *directly* on the finite difference of interest can indeed approximate it; what fails is differentiating the model trained on levels.

8 The Mexican regulatory framework

CVA is neither an academic exercise nor an optional practice. In Mexico it rests on four references worth distinguishing.

IFRS 13 (fair value). Establishes that fair value must reflect the assumptions market participants would use in pricing, including those about credit risk. It follows that the counterparty credit adjustment is part of the measurement.

NIF C-10 and Annex 33 of the CUB (CNBV). Criterion B-5, on derivatives and hedging transactions, refers to NIF C-10 for the corresponding accounting treatment.

Banco de México's XVA report. Requires periodic reporting of XVA adjustments, counterparty by counterparty, with legal basis in the Banco de México Act and the Credit Institutions Act.

Basel framework (BIS, MAR 50). Defines regulatory capital for CVA risk.

Two different things that share a name

Accounting CVA must not be confused with regulatory capital for CVA risk. The first is a valuation adjustment under the measure $\mathbb{Q}$, reported in Banxico's form. The second **excludes** own credit and applies supervisory risk weights to sensitivities to produce a capital requirement: it takes $\mathbb{Q}$ -measure sensitivities as an input, but it is not the valuation adjustment.

One piece of local context is worth adding. Since November 2024 the Mexican market has migrated to TIIE de Fondeo as the reference rate for derivatives, replacing 28-day TIIE. That change rebuilt the curves these transactions are priced off, so the valuation and exposure infrastructure had to be rebuilt as well. It is one reason there is demand for people able to work in this area.

9 Summary

1. Two transactions with identical cash flows are not worth the same if the counterparties differ in credit quality. That difference is the CVA.
2. CVA decomposes into severity, exposure and probability. The probability is read off the CDS market because CVA is a price, not a forecast.
3. Exposure requires simulating the future. PCA of the TIIE de Fondeo curve shows that two factors —level and slope— explain close to 90% of the variance, which justifies the G2++.
4. Expected exposure is hump-shaped, peaking around $T/3$: the risk is not where intuition puts it.
5. CVA and DVA are two readings of the same simulated distribution, split at zero. They net, and the sign of the adjustment depends on which party is riskier.
6. At portfolio scale and with intraday needs, the Monte Carlo engine is not enough. A surrogate trained on labels from the engine itself answers in microseconds without replacing it: the engine still validates and signs.
7. The right model depends on the question: trees for levels, smooth models for sensitivities obtained by differentiation.

10 Further reading

For those who want to go deeper, these references are ordered from most accessible to most technical.

- Gregory, J. (2012). *Counterparty Credit Risk and Credit Value Adjustment*. Wiley. The industry reference; readable without advanced stochastic calculus.

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- Green, A. (2015). *XVA: Credit, Funding and Capital Valuation Adjustments*. Wiley. Covers the full family of valuation adjustments.
 - Brigo, D. and Mercurio, F. (2006). *Interest Rate Models — Theory and Practice*. Springer. The source of the G2++ and of the closed-form results used here.
 - Litterman, R. and Scheinkman, J. (1991). Common factors affecting bond returns. *The Journal of Fixed Income*. The paper that established the level, slope and curvature interpretation.